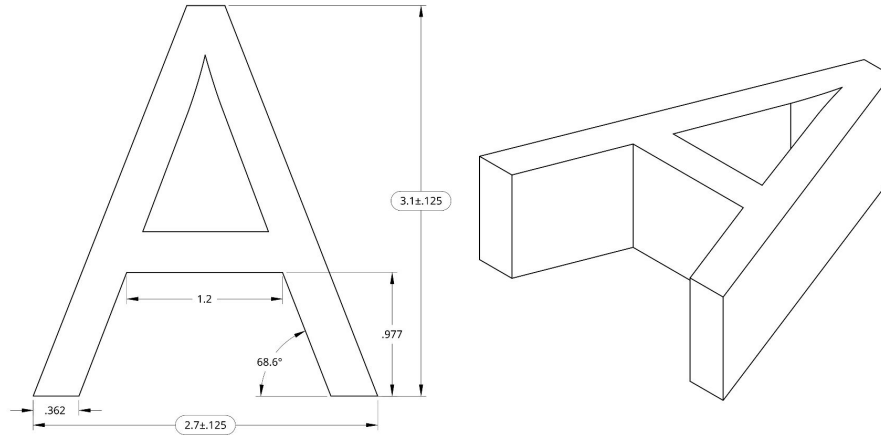


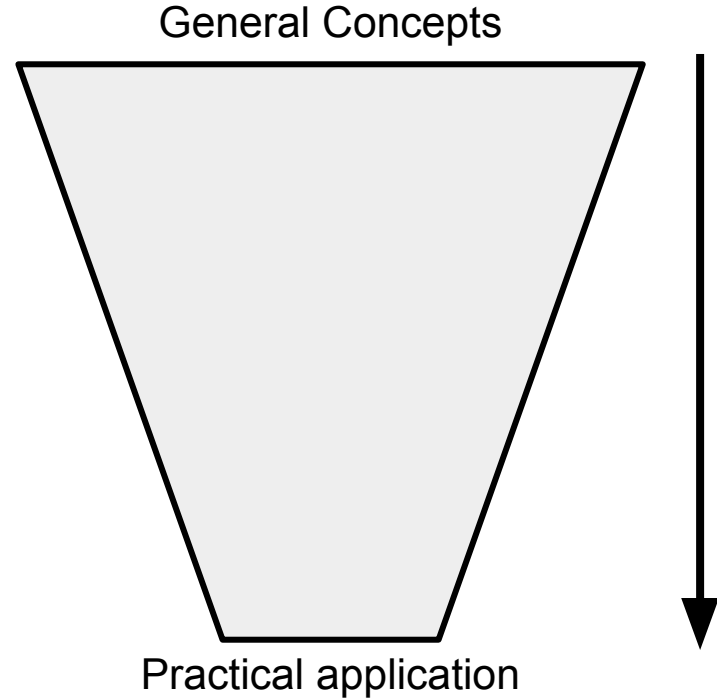
The Definition of Quality

Clarity for modern business & manufacturing



Overview

- Why define quality?
- Subjective vs objective quality
- The traditional approach
- The Taguchi loss function
- Two types of variation
- A practical example
- Questions
- Additional resources
- Appendix



Why define quality?

A term that means nothing in particular means nothing at all.

— Stephen Few
Big Data, Big Dupe

Some examples of cooperation

- Date and time of day
- Sizes of batteries
- Red and green traffic lights
- The metric system
- Ratio of focal length to a lens diameter refers to a wavelength of 546 nanometers
- Railway gauge
- Standard voltage across all of North America

Subjective vs Objective Quality

Subjective Quality

Objective Quality



What do **I** think?

This part is **high** quality

The part has a **good** finish

The part has **acceptable** flatness

What can **we** measure?

The measured **diameter** is 25.03 mm

The **surface roughness** must be $\leq 0.8 \mu\text{m}$

The measured **flatness** is 0.05 mm

To me, quality is consistently meeting requirements while making sure the process is capable of doing it again tomorrow. It's not just catching a bad part at final inspection. It's building the process so the bad part is less likely to happen in the first place.

— Aerospace & Defence Quality Manager

Quality is what the customer wants and it varies.

— Food & Beverage Plant Manager

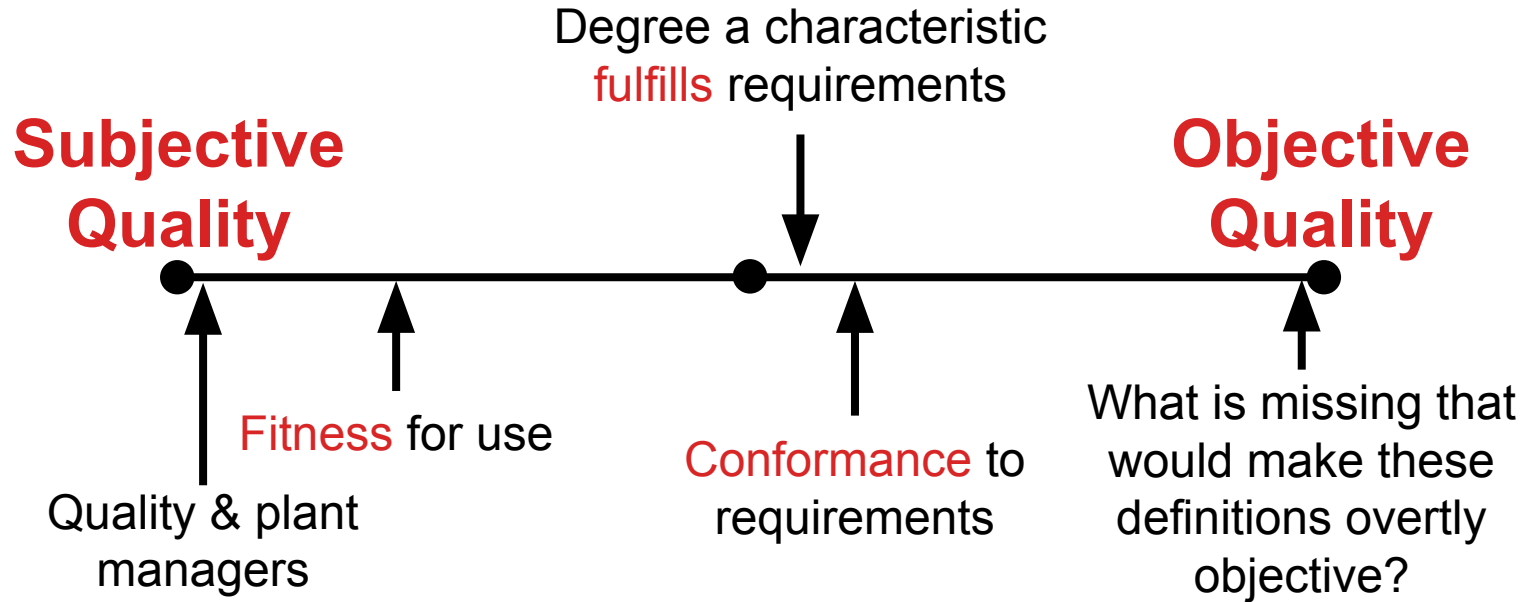
Some common definitions of quality

Fitness for use (Juran)

Conformance to requirements (Crosby)

The degree to which a set of inherent characteristics **fulfills** requirements (ISO 9000)

Where do the prior definitions fall along the continuum?



What is variation?

The degree to which the sequential outputs of a process or system differ.

Take the task of drawing a sequence of capital letter As.



We are limited in doing what we want to do; that to do what we set out to do, even in so simple a thing as making a's that are alike, requires almost infinite knowledge compared with that which we now possess. It follows, therefore, since we are thus willing to accept as axiomatic that we cannot do what we want to do and cannot hope to understand why we cannot, that we must also accept as axiomatic that a controlled quality will not be a constant quality. Instead, a controlled quality will be a variable quality.

— Walter A. Shewhart
Economic Control of Quality of Manufactured Product

The traditional approach

Craft production

Low volume

Manual

Skilled craftsmen

Not interchangeable parts

Mass production

High volume

Automated

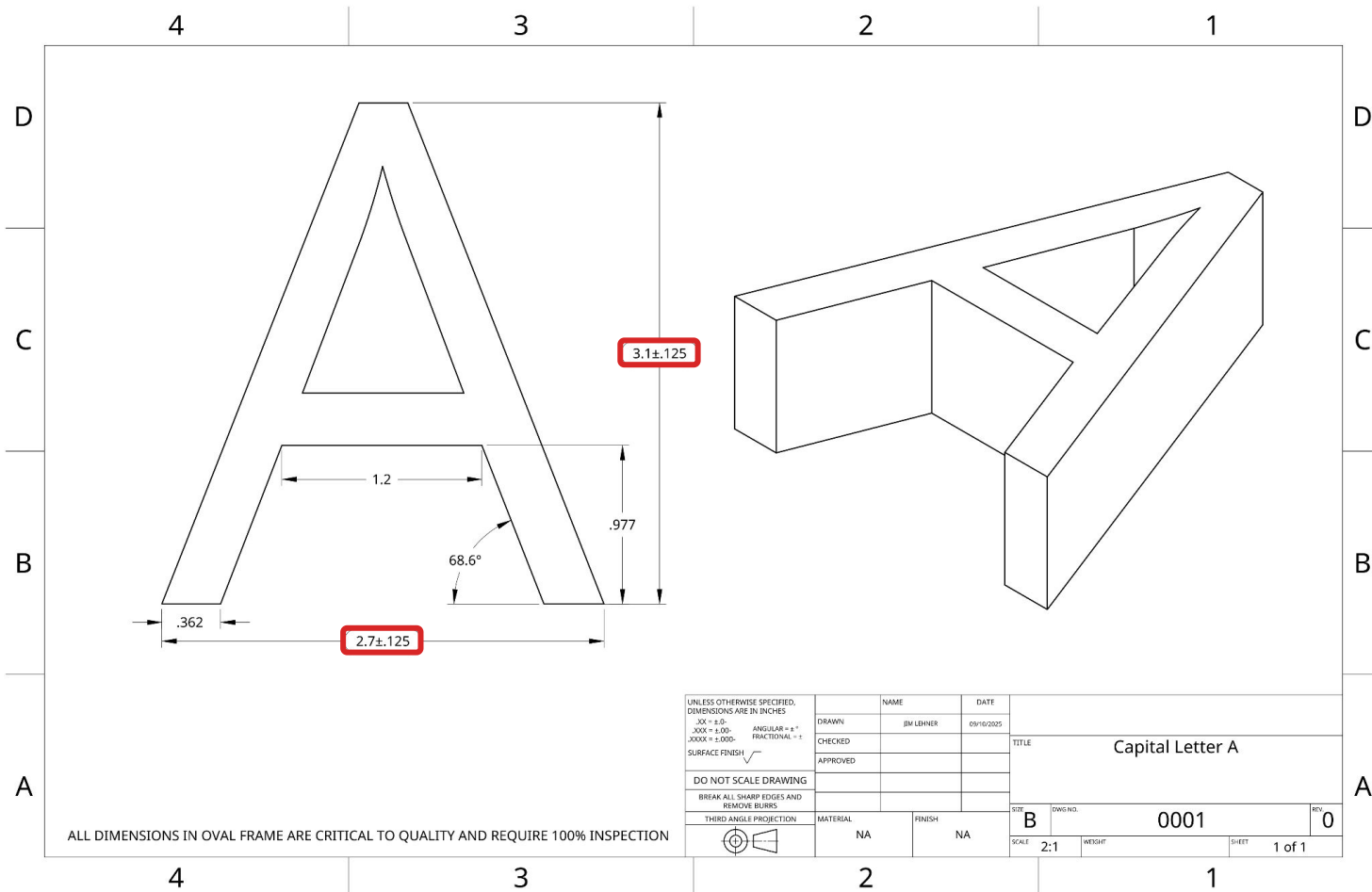
Unskilled labor

Interchangeable parts

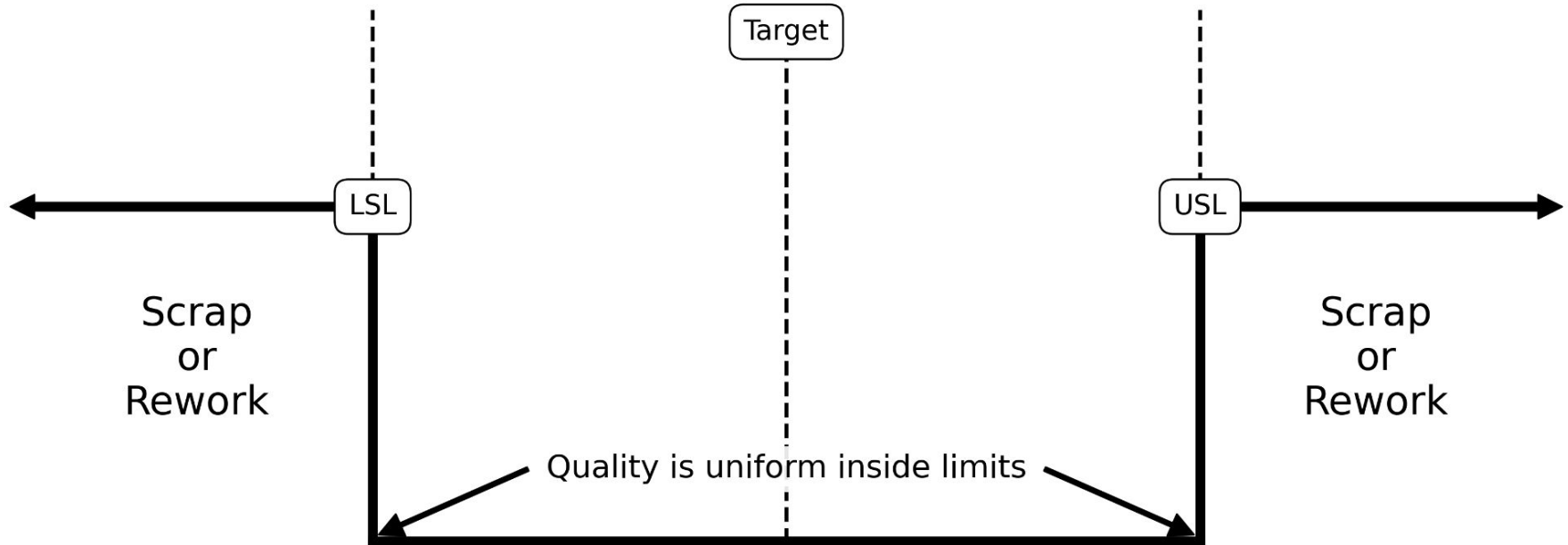
What are specifications?

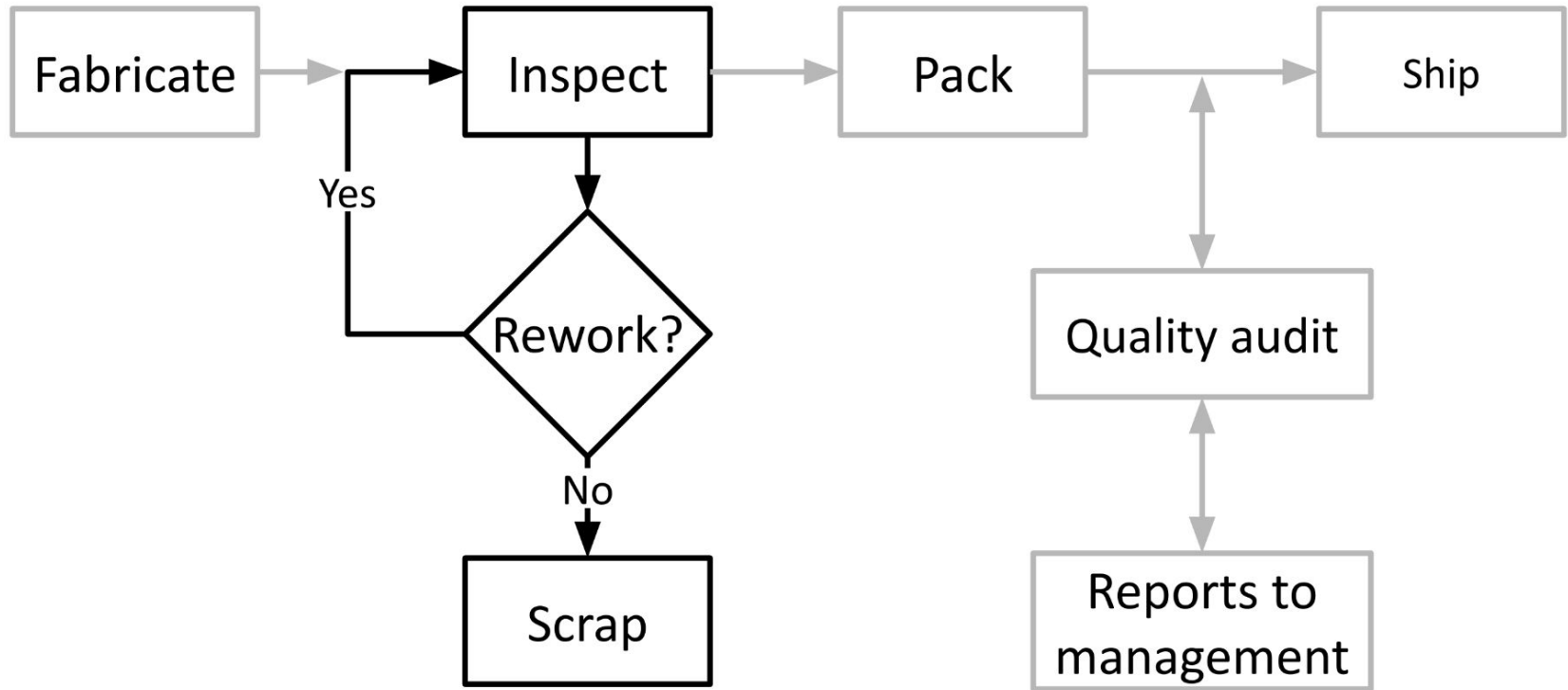
The extent to which parts can differ and still satisfy the customer need.





Rectangular loss function







The Taguchi loss function

What is the definition of world-class quality?

On-target with minimum variance.

World-class quality can be approximated with a quadratic loss function of the form:

$$L(x) = K (x - T)^2$$

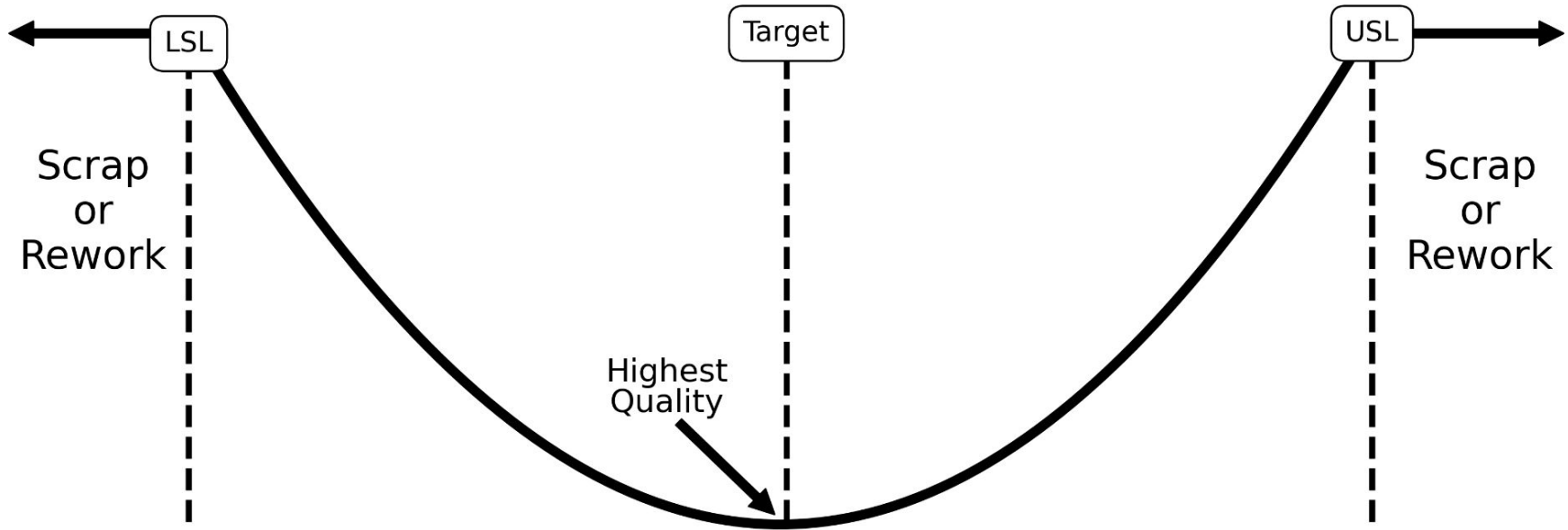
$L(x)$ is the loss function

K is a numeric constant

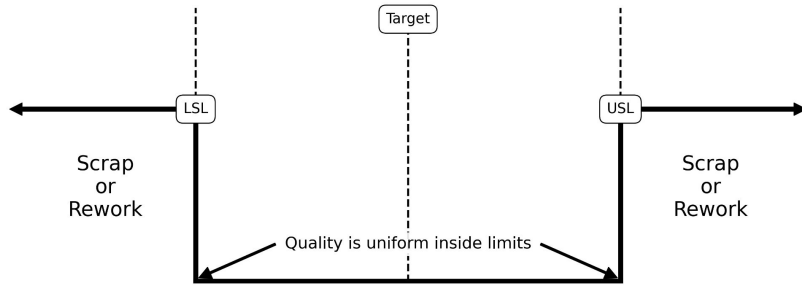
x is the value of the performance or quality characteristic

T is the target value

Taguchi loss function



Square loss function



Low fidelity model of loss
Loss occurs **abruptly**
Promotes **reactionary efforts**

Taguchi loss function



High fidelity model of loss
Loss occurs **incrementally**
Promotes **continual improvement**

Two types of variation

Common causes

Produce **routine** variation

Many in number

Small effect on process behavior

Predictable and consistent

Controlled

Assignable causes

Produce **exceptional** variation

Few in number

Large effect on process behavior

Unpredictable and inconsistent

Uncontrolled

Common causes

Poor lighting

Vibration

Humidity not suited to the process

Uncomfortable working conditions

Assignable causes

Produce **exceptional** variation

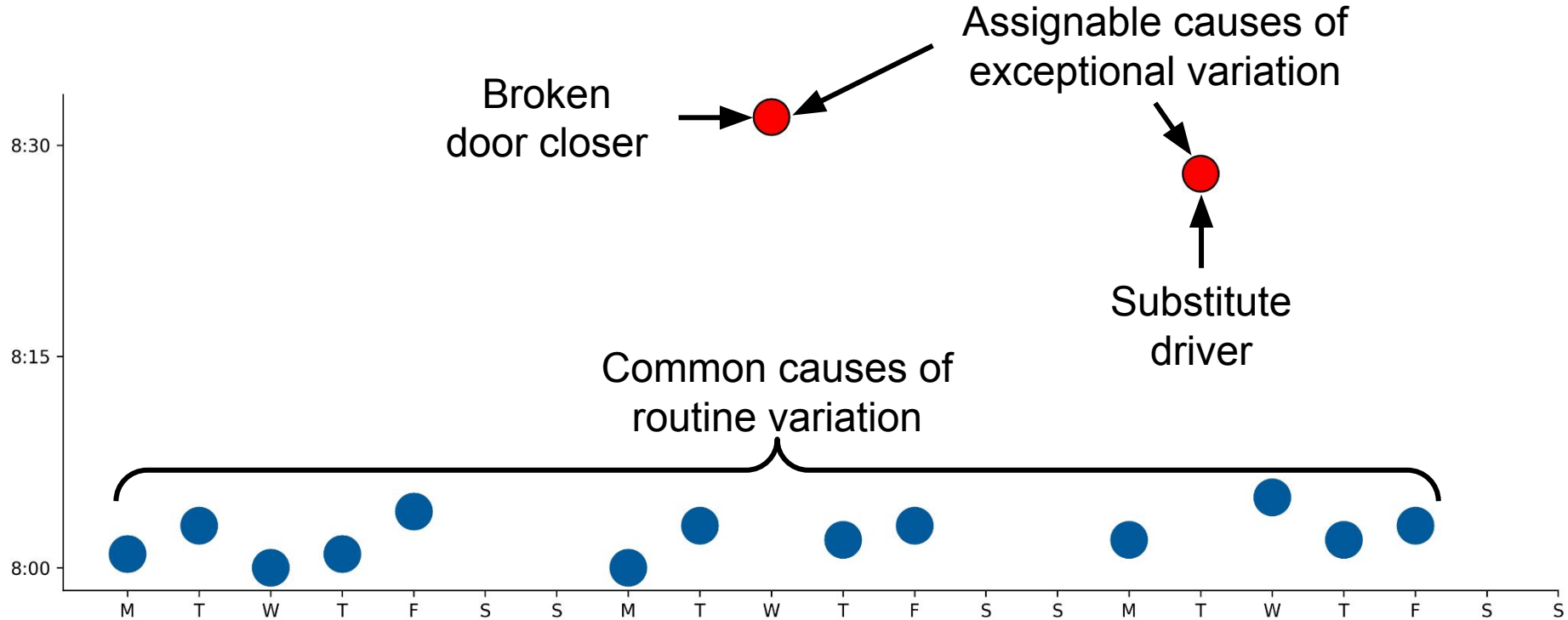
Few in number

Large effect on process behavior

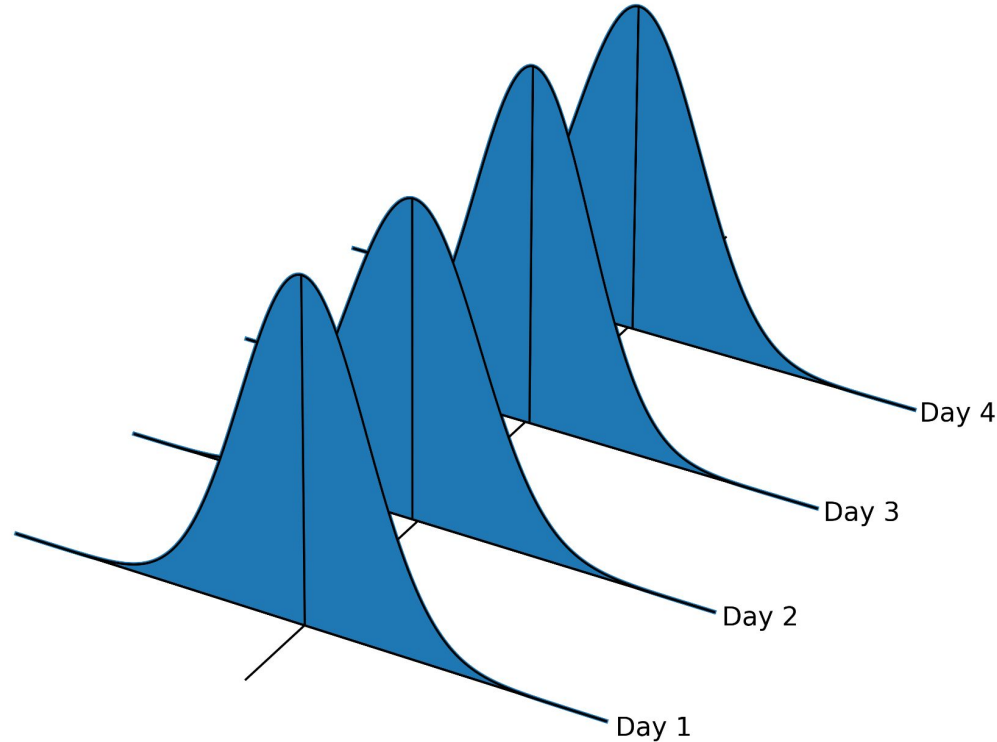
Unpredictable and inconsistent

Uncontrolled

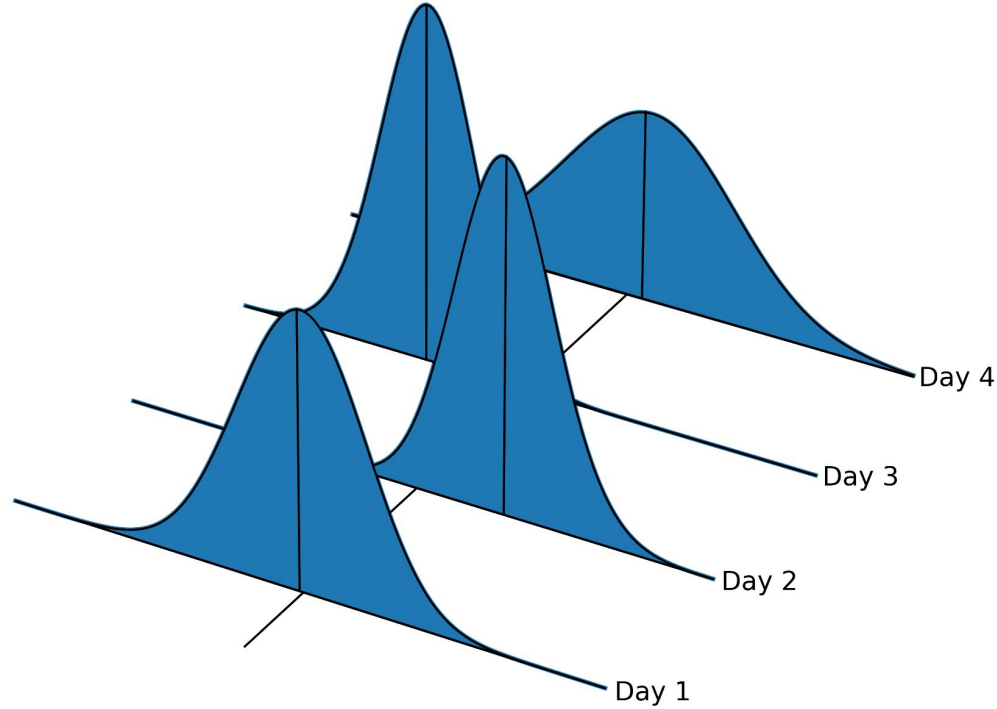
Arrival of the school bus



What can be said about these distributions?

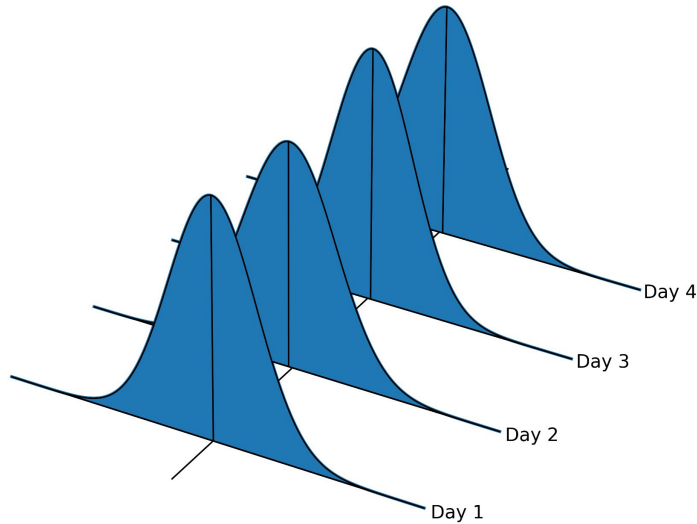


What can be said about these distributions?

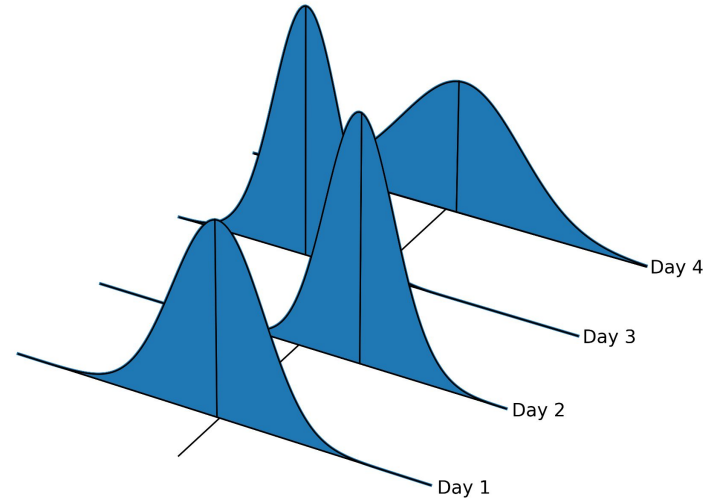


How do we know if a process is predictable or unpredictable?

Predictable process



Unpredictable process



A practical example

204 electrical resistance measurements

5,045	4,635	4,700	4,650	4,640	3,940	4,570	4,560	4,450	4,500	5,075	4,500
4,350	5,100	4,600	4,170	4,335	3,700	4,570	3,075	4,450	4,770	4,925	4,850
4,350	5,450	4,110	4,255	5,000	3,650	4,855	2,965	4,850	5,150	5,075	4,930
3,975	4,635	4,410	4,170	4,615	4,445	4,160	4,080	4,450	4,850	4,925	4,700
4,290	4,720	4,180	4,375	4,215	4,000	4,325	4,080	3,635	4,700	5,250	4,890
4,430	4,810	4,790	4,175	4,275	4,845	4,125	4,425	3,635	5,000	4,915	4,625
4,485	4,565	4,790	4,550	4,275	5,000	4,100	4,300	3,635	5,000	5,600	4,425
4,285	4,410	4,340	4,450	5,000	4,560	4,340	4,430	3,900	5,000	5,075	4,135
3,980	4,065	4,895	2,855	4,615	4,700	4,575	4,840	4,340	4,700	4,450	4,190
3,925	4,565	5,750	2,920	4,735	4,310	3,875	4,840	4,340	4,500	4,215	4,080
3,645	5,190	4,740	4,375	4,215	4,310	4,050	4,310	3,665	4,840	4,325	3,690
3,760	4,725	5,000	4,375	4,700	5,000	4,050	4,185	3,775	5,075	4,665	5,050
3,300	4,640	4,895	4,355	4,700	4,575	4,685	4,570	5,000	5,000	4,615	4,625
3,685	4,640	4,255	4,090	4,700	4,700	4,685	4,700	4,850	4,770	4,615	5,150
3,463	4,895	4,170	5,000	4,700	4,430	4,430	4,440	4,775	4,570	4,500	5,250
5,200	4,790	3,850	4,335	4,095	4,850	4,300	4,850	4,500	4,925	4,765	5,000
5,100	4,845	4,445	5,000	4,095	4,850	4,690	4,125	4,770	4,775	4,500	5,000



Given the summary statistics, what next?

5,045	4,635	4,700	4,650	4,640	3,940	4,570	4,560	4,450	4,500	5,075	4,500
4,350	5,100	4,600	4,170	4,335	3,700	4,570	3,075	4,450	4,770	4,925	4,850
4,350	5,450	4,110	4,255	5,000	3,650	4,855	2,965	4,850	5,150	5,075	4,930
3,975	4,635	4,410	4,170	4,615	4,445	4,160	4,080	4,450	4,850	4,925	4,700
4,290	4,720	4,180	4,375	4,215	4,000	4,325	4,080	3,635	4,700	5,250	4,890
4,430	4,810	4,790	4,175	4,275	4,845	4,125	4,425	3,635	5,000	4,915	4,625
4,485	4,565	4,790	4,550	4,275	5,000	4,100	4,300	3,635	5,000	5,600	4,425
4,285	4,410	4,340	4,450	5,000	4,560	4,340	4,430	3,900	5,000	5,075	4,135
3,980	4,065	4,895	2,855	4,615	4,700	4,575	4,840	4,340	4,700	4,450	4,190
3,925	4,565	5,750	2,920	4,735	4,310	3,875	4,840	4,340	4,500	4,215	4,080
3,645	5,190	4,740	4,375	4,215	4,310	4,050	4,310	3,665	4,840	4,325	3,690
3,760	4,725	5,000	4,375	4,700	5,000	4,050	4,185	3,775	5,075	4,665	5,050
3,300	4,640	4,895	4,355	4,700	4,575	4,685	4,570	5,000	5,000	4,615	4,625
3,685	4,640	4,255	4,090	4,700	4,700	4,685	4,700	4,850	4,770	4,615	5,150
3,463	4,895	4,170	5,000	4,700	4,430	4,430	4,440	4,775	4,570	4,500	5,250
5,200	4,790	3,850	4,335	4,095	4,850	4,300	4,850	4,500	4,925	4,765	5,000
5,100	4,845	4,445	5,000	4,095	4,850	4,690	4,125	4,770	4,775	4,500	5,000

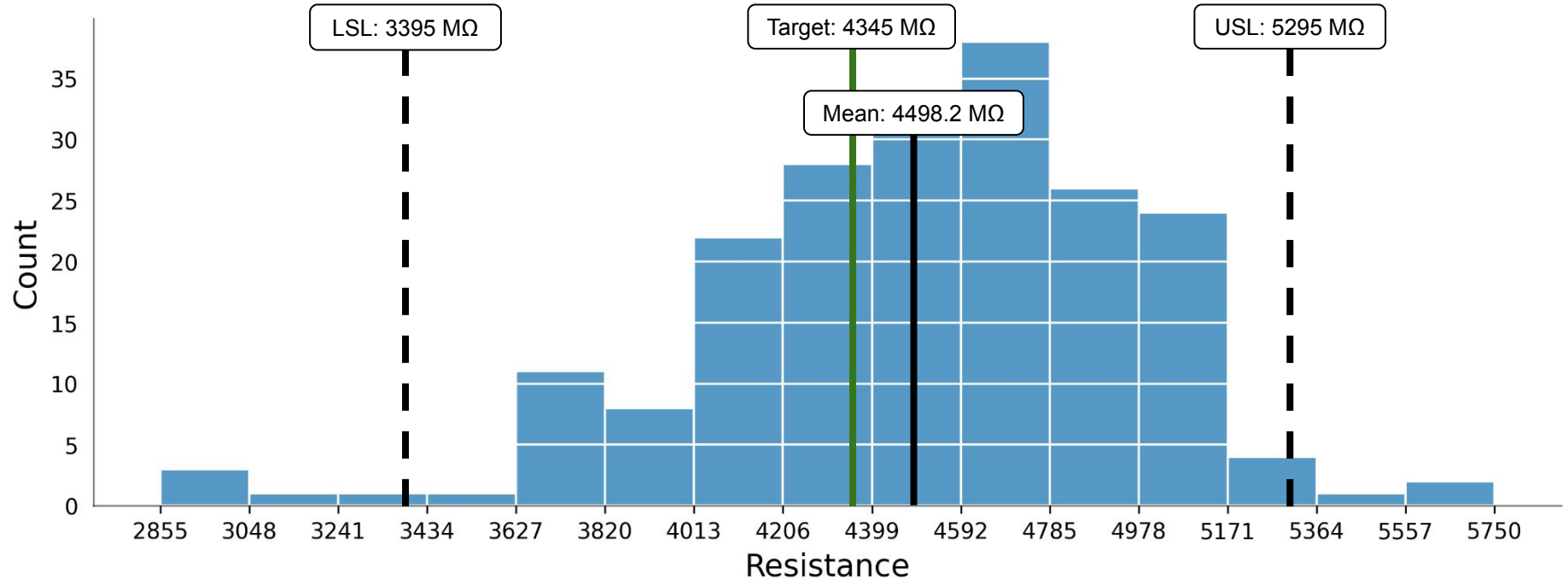
Mean:
4498.2

Median:
4565

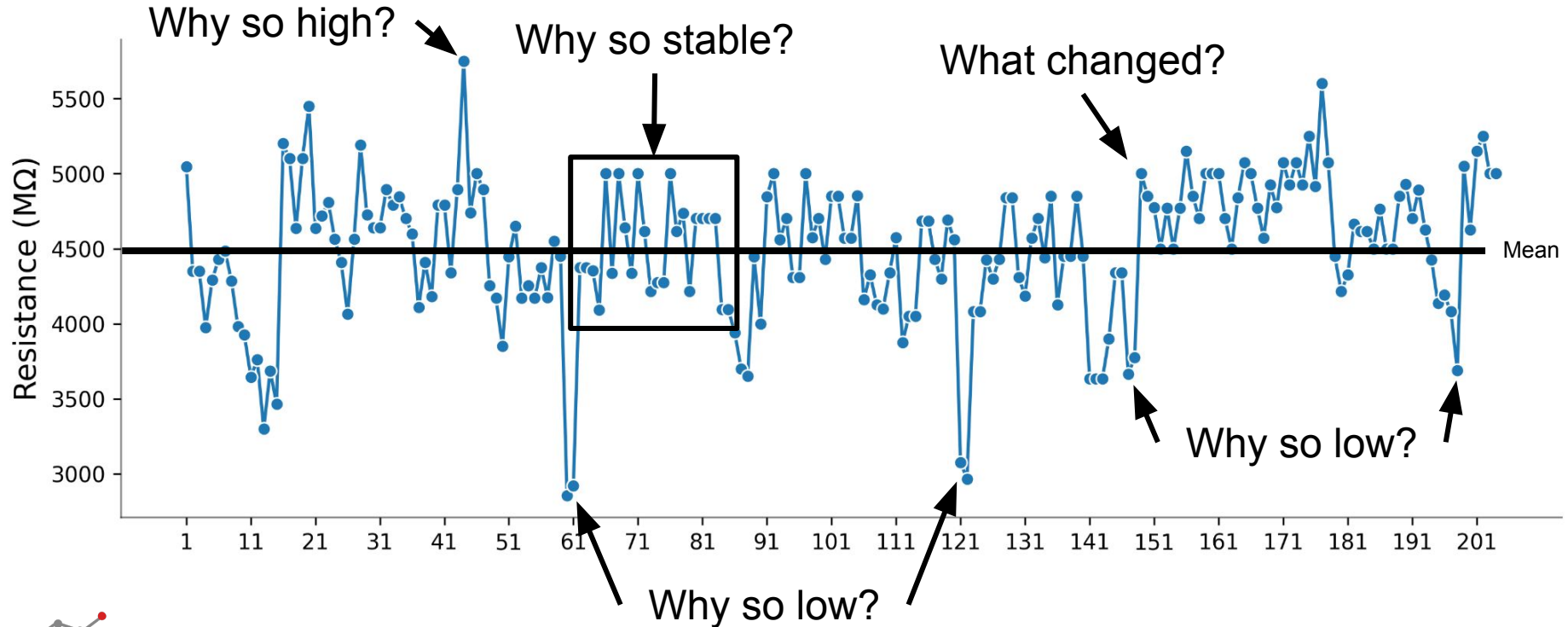
Range:
2,895

Standard deviation:
466.4

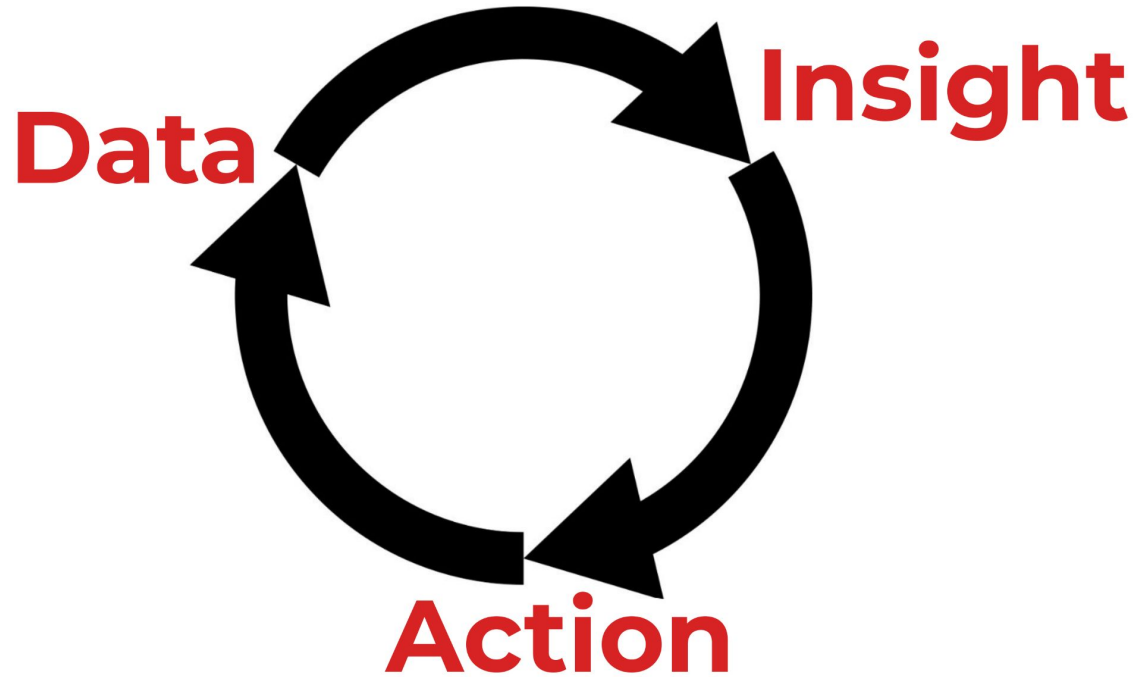
Given the histogram, what next?



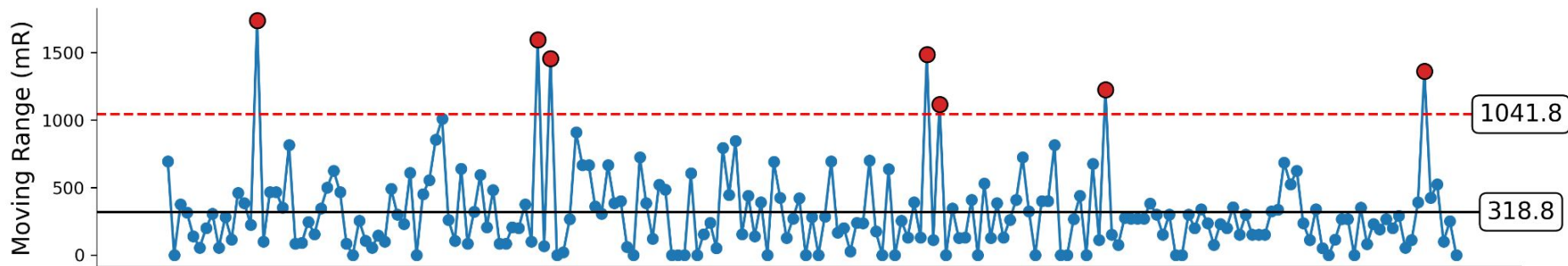
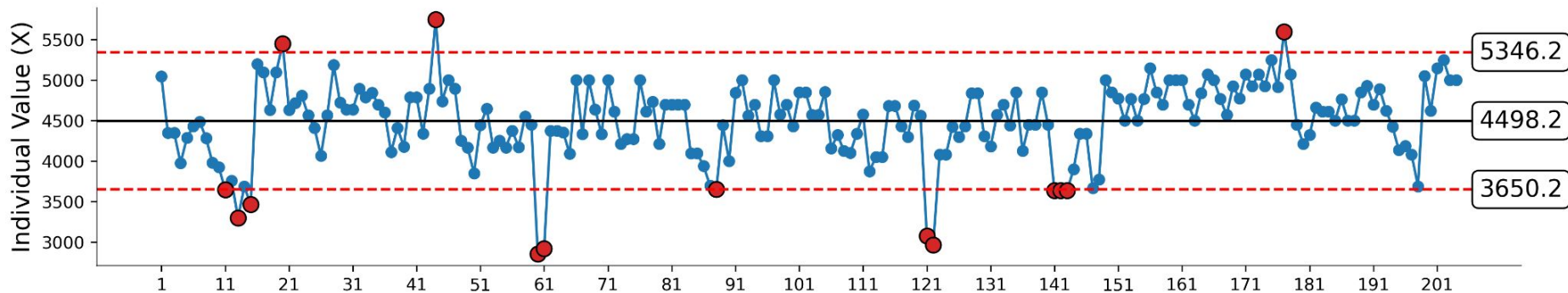
What about a time series?



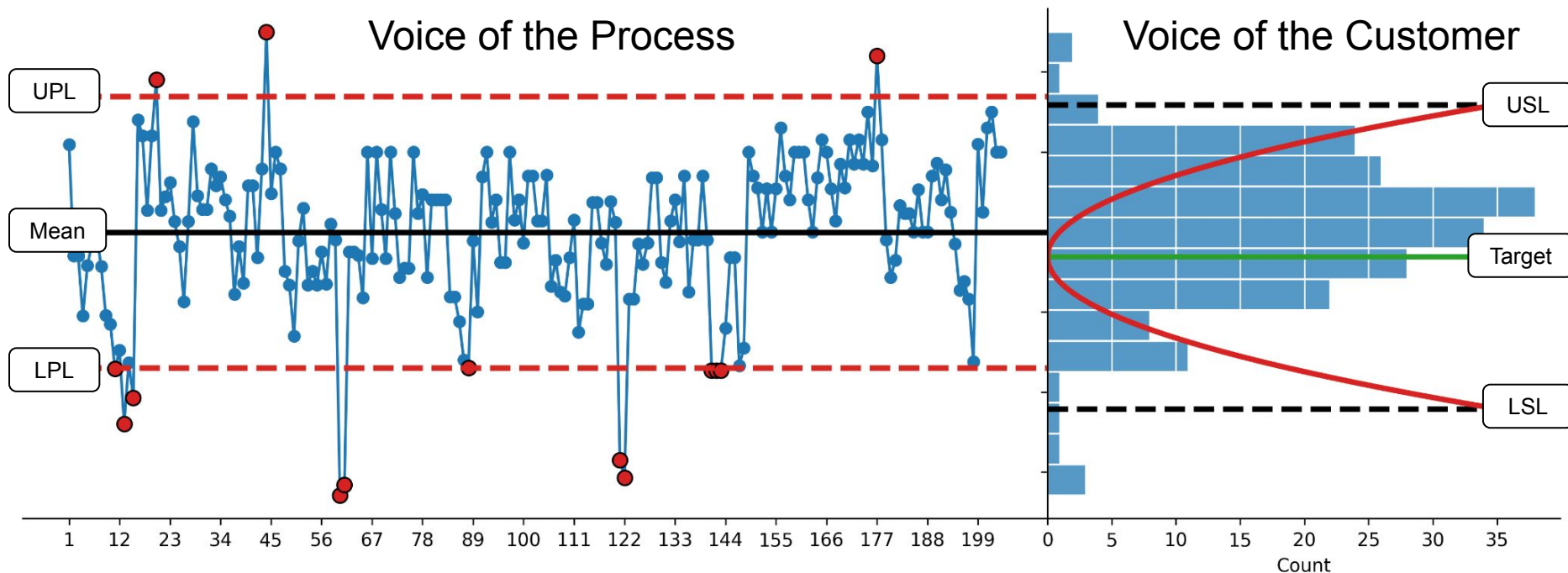
The DIA Cycle



XmR chart of 204 resistance measurements



Combo chart of resistance measurements



So what is the definition of quality?

On-target with minimum variance

Questions

Additional resources

- A Crash Course in Variation (store.brokenquality.com/b/crash-course)
- XmR Chart-It Web App (xmrchart.streamlit.app)
- Taguchi Loss Web App (taguchi-loss.streamlit.app)
- The Broken Quality Initiative (brokenquality.com)
- The Broken Quality Initiative Store (store.brokenquality.com)
- The Math Matters Newsletter (themathmatters.substack.com)

Appendix

Process limits for an XmR chart:

$$\text{UPL} = \text{Average} + (2.660 \cdot \text{Ave. mR})$$

$$\text{LPL} = \text{Average} - (2.660 \cdot \text{Ave. mR})$$

$$\text{URL} = 3.268 \cdot \text{Ave. mR}$$